

Non-Technical Report

In baseball performance training, lower-body mechanics and force output are closely monitored for their relationship to throwing velocity. Lower body strength and force output are key factors in training pitchers and building up the body to handle the forces demanded in throwing. At Kinetic Performance Institute in California, we collect countermovement jump (CMJ) data as part of our athlete monitoring system. This report explores a recent analysis of CMJ test results and how various force plate metrics relate to throwing velocity. The data comes from one of three force plate jump variations we use every day. In future work, we plan to expand the analysis to include all jump types to build a more complete predictive profile. This initial model offers insights that can support coaching and training decisions. The model produced a final RMSE (root mean squared error) of 6.31, meaning it predicts throwing velocity within about ± 6.3 mph on average. Although this is not an acceptable level of accuracy for true assessments, it gives us other valuable information; the strength of certain metrics within the jump.

Out of more than 79 potential predictors, we will cover the 7 most important for model prediction power. These metrics had the most impact on our model's ability to predict velocity, giving us a ranking. The most important were flight time and average propulsive power, both of which relate to how much and how efficiently an athlete pushes off the ground. These measures reflect not just strength but coordinated force output, central to both jumping and pitching. Next in importance were propulsive net impulse and positive net impulse, which represent the total force applied over time during the upward (propulsive) phase. These findings support the use of time-constrained plyometric work in combination with heavy full-body loading with weightlifting, building both the base of strength to produce enough force against the ground and the speed to move the load fast. Two braking phase metrics also made the top seven: stiffness and average braking power. These relate to how well an athlete absorbs force and rapidly transitions to upward motion. In practice, it emphasizes drop-based plyometrics with overcoming isometrics, focusing on body control and elasticity.

Although this analysis has vast room for improvement in model prediction power, it does give us applicable training insights that can guide coaches when interpreting force plate metrics for their athletes. Focusing on the change of momentum and force capacity during acceleration is key; they should be the main goal in the weight room for young athletes looking to improve velocity. Moving fast through these changes seems to be very important; we need to jump and move quickly in our training. These are just some examples of what we can take from this analysis. However, in the future, including some history of injury could be helpful in the context of prediction. We would also revamp the data and use a broader scope of athletes (~500-700). There is a possibility of also including other jump metrics. As the counter movement jump is strong in assessing propulsive and braking forces, others, like the multi-rebound, give a better perspective on elasticity and prolonged force output. The future is bright, and with more time and data, the potential for informed training decisions only grows.